

Collision Frequency Trends for Various Vehicle Types in the Greater Toronto Area

INF2167 Project - Group 7

Stella Gregorski Yiyang Qin

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Introduction

Why is traffic system important?

- Economic and social impacts
- Urban areas face increasing collision risks
- Data-driven analysis supports evidence-based planning

Motivation

What temporal and spatial patterns exist in Toronto traffic collisions, and how do collision risks vary across different travel modes?

- Study on Wujiang, China ([Cheng et al. 2018](#)) discovered high-risk spot at the connection with a major city, which is associated with high traffic volumes, ongoing construction projects, and high proportion of large trucks.
- Scientists in Iran ([Yousefzadeh-Chabok et al. 2016](#)) conducted a time series analysis in Zanjan Province for predicting the future morality in traffic, and the models shows a descending trend in traffic morality, associated with stricter traffic enforcement and education as time passed.
- Investigation on Toronto collision patterns supports evidence-based transportation planning and resource allocation

Data

- The Toronto Police Service maintains data on traffic collisions they are called to respond to
- This data includes attributes such as:
 - **Month, Weekday, and Hour** when the collision took place
 - **Neighbourhood** where the collision took place
 - Binaries for whether five different types of vehicle were involved in the collision
 - These include **Automobile, Passenger Vehicle (public transportation), Motorcycle, Bicycle, and Pedestrian**
- We obtained this data from (**Toronto Police Service 2023**) for the time frame 2020-2026

Methods

- This project used the R programming language ([R Core Team 2026](#)) and the tidyverse library ([Wickham et al. 2019](#)) to do reproducible data analysis.
- Two primary steps were used to determine spatial and temporal trends in Toronto traffic collisions:
 1. Exploratory Data Analysis and Statistical Visualization
 2. Linear Regression Statistical Modeling

Findings

- Collisions for all vehicle types are most frequent in the afternoon (from 13:00 - 17:00)
- Collisions for motorized vehicles are more frequent in the winter, while bicycles and pedestrians are in more collisions during the summer
- While spatial and temporal trends can be observed visually, they are not particularly well-correlated
 - Residuals are centered around zero, but have observable patterns and a non-constant spread
 - Adding vehicle type to the model worsened the correlation

Implications

- Rush hour and winter weather are primary indicators of collision frequency
- We can temporally and spatially model collision risk, similar to prior research
- Until we have more data on non-automobile collisions, trends in these types of accidents may remain less modelable

Conclusion

- Key takeaways from the project
- Future research ideas

References

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